

# BLM224 Elektronik

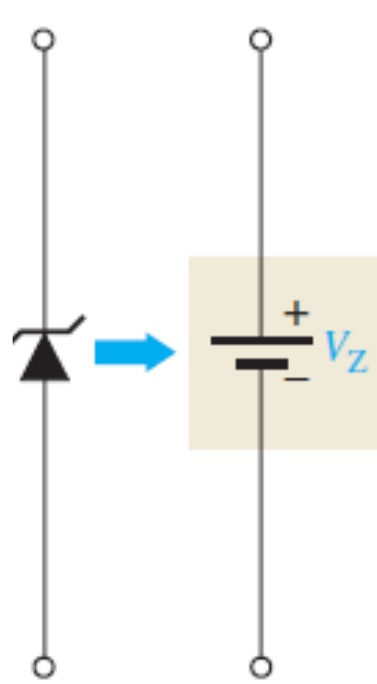
Hafta 4

Zener Diyot ve Örnek Çözümler

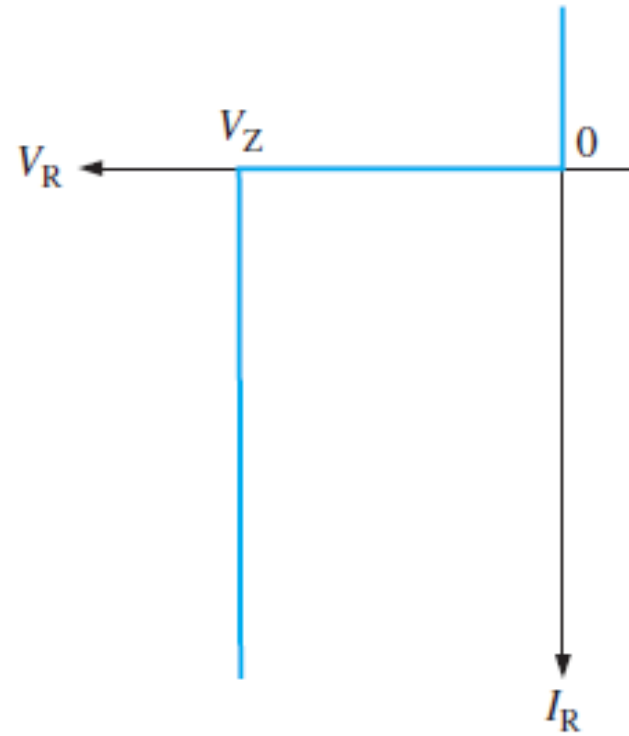
Karabük Üniversitesi

Bilgisayar Mühendisliği

# Zener Eşdeğer Devresi

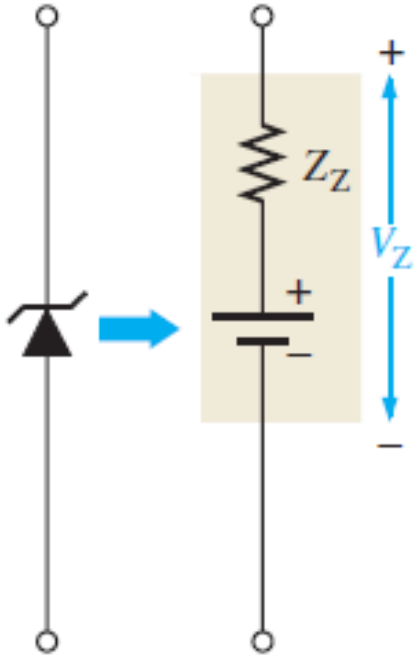


İdeal model

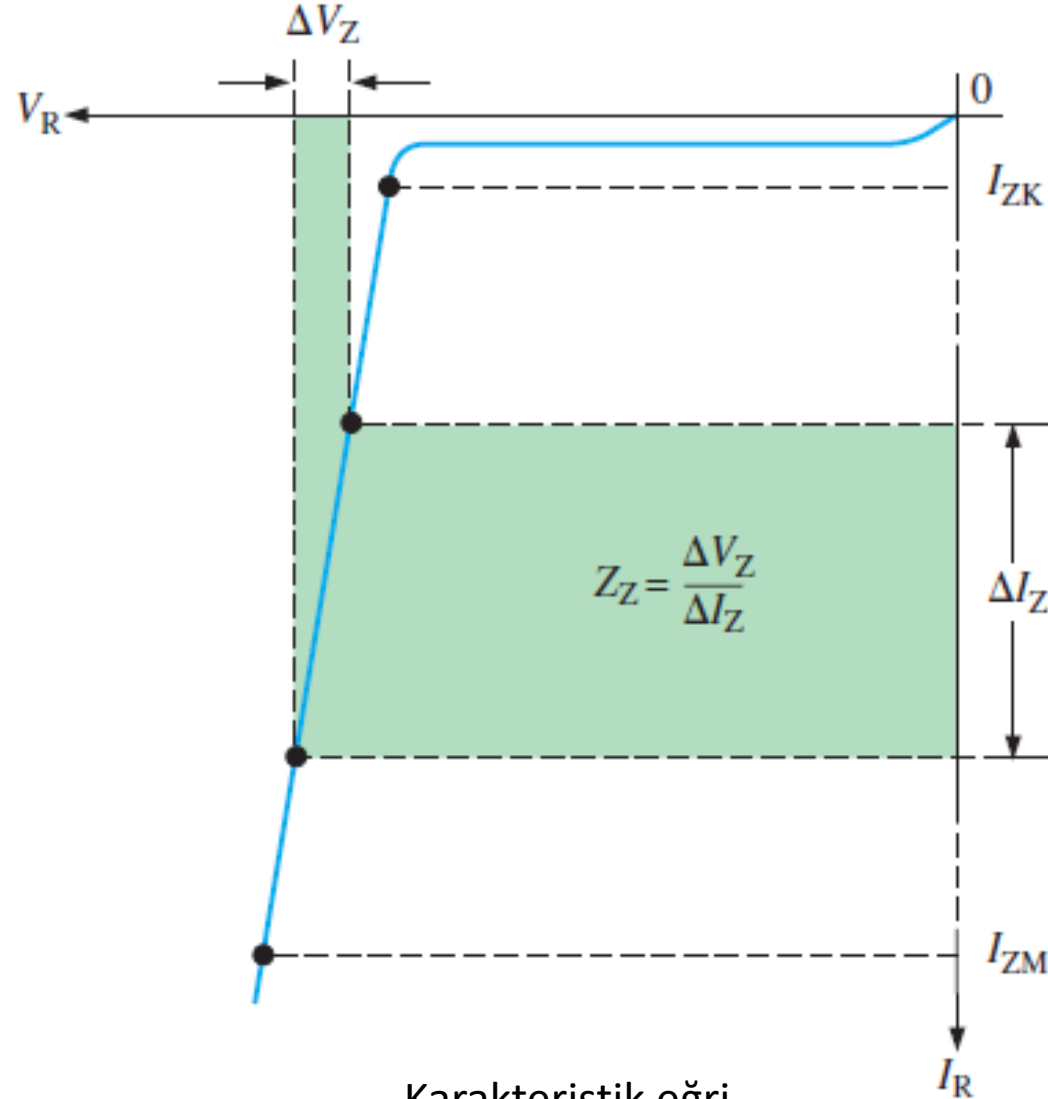


İdeal karakteristik

# Zener Eşdeğer Devresi



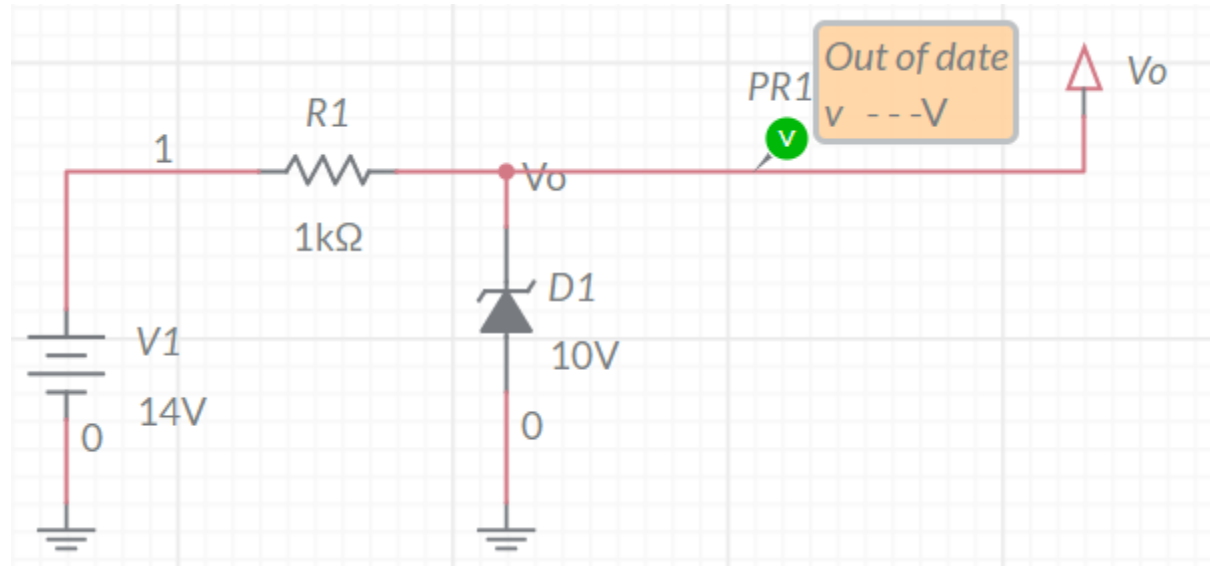
Pratik model



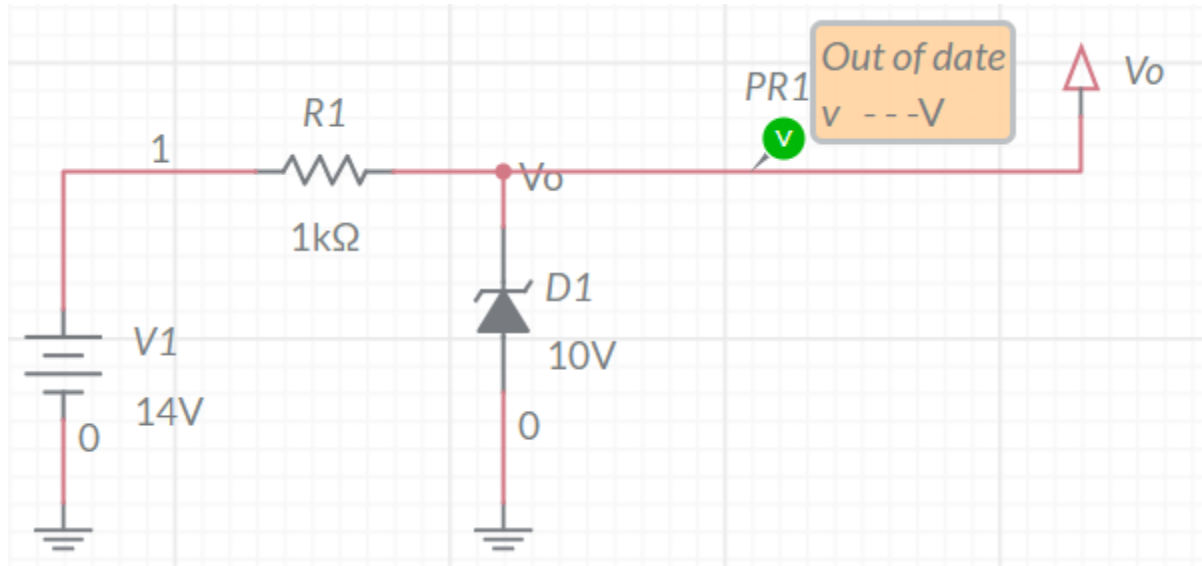
Karakteristik eğri

$$R_z = Z_z = \frac{\Delta V_z}{\Delta I_z}$$

# Örnek



# Örnek



$$V_o = V - V_R = V_Z$$

$$V_o = V - RI = V_Z$$

$$V_o = V_Z = 10V$$

$$I = \frac{14 - 10}{1k} = 0.004 A$$

$$I = \frac{14 - 10}{500} = 0.008 A$$

$$I = \frac{14 - 10}{250} = 0.016 A$$

R değıştikçe akım değışiyor fakat, çıkış gerilimi sabit kalıyor. \*\* $R_Z$  ihmal edilmezse çıkış azda olsa değışir.

Örnek:

1N1736A zener diyodunun katalogundan 37mA'lık test akımında zener empedansı(direnci)  $Z_z = 3.5\Omega$  ve zener gerilimi  $V_z = 6.8V$  okunmuştur.

- a) Zenerden geçen akım 50mA iken zenerin gerilimini bulunuz.
- b) Zenerden geçen akım 25mA iken zenerin gerilimini bulunuz.

Çözüm:

$$\text{a) } \Delta I_Z = I_Z - 37 \text{ mA} = 50 - 37 = 13 \text{ mA}$$

$$\Delta V_Z = Z_Z \times \Delta I_Z = 3.5 \Omega \times 13 \text{ mA} = 45.5 \text{ mV}$$

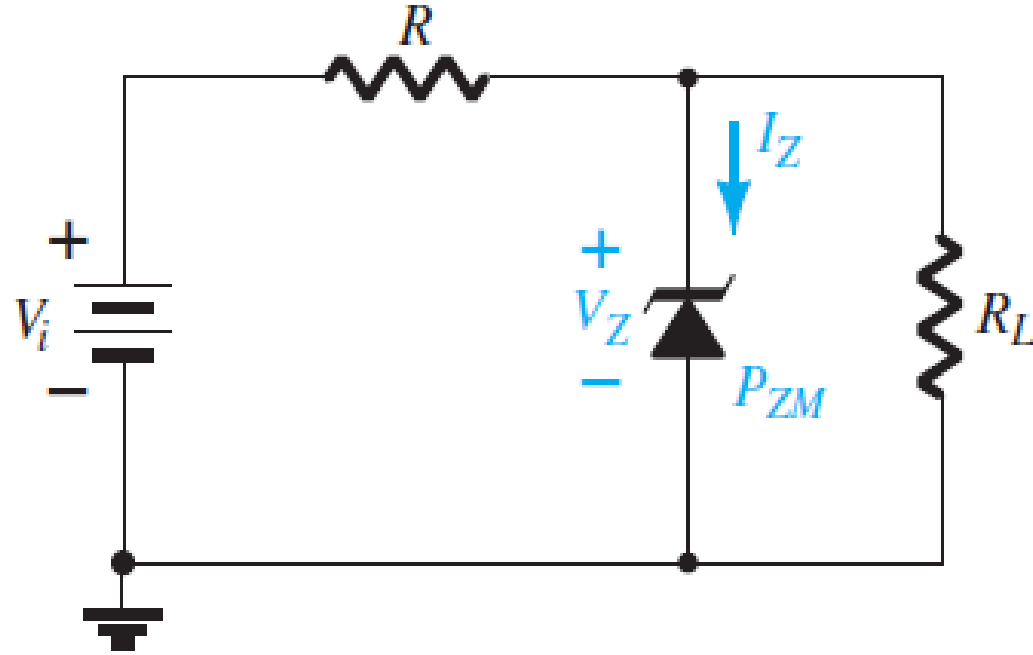
$$V_Z = 6.8 \text{ V} + \Delta V_Z = 6.8 \text{ V} + 45.5 \text{ mV} = \mathbf{6.85 \text{ V}}$$

$$\text{b) } \Delta I_Z = I_Z - 37 \text{ mA} = 25 - 37 = -12 \text{ mA}$$

$$\Delta V_Z = Z_Z \times \Delta I_Z = 3.5 \Omega \times (-12 \text{ mA}) = -42 \text{ mV}$$

$$V_Z = 6.8 \text{ V} + \Delta V_Z = 6.8 \text{ V} + (-42.5 \text{ mV}) = \mathbf{6.76 \text{ V}}$$

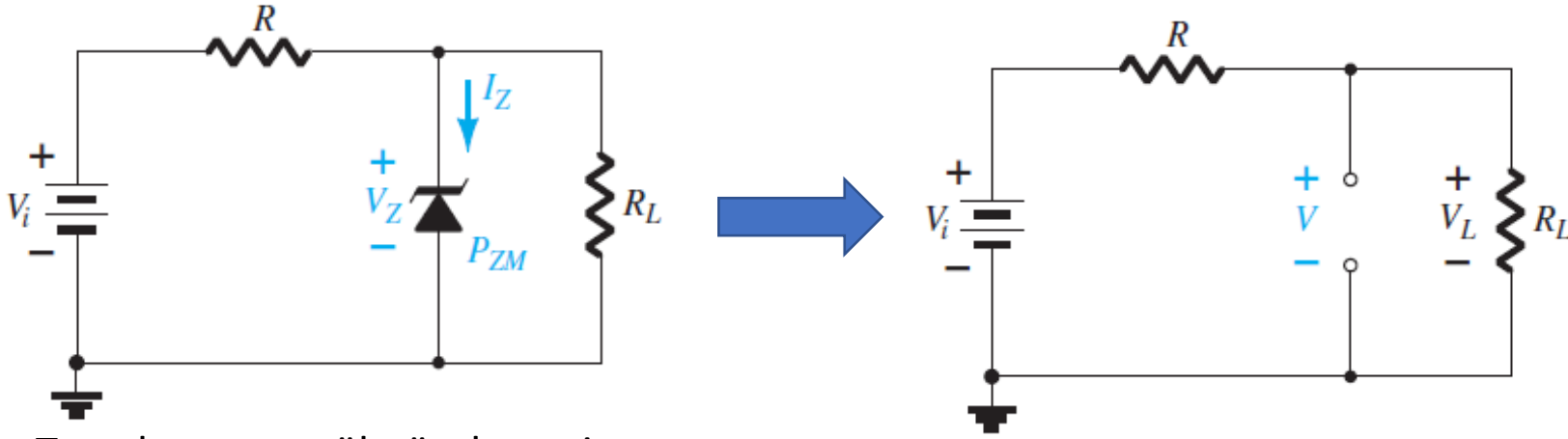
# Zener Diyotun Regülatör Olarak Kullanılması



Temel zener regülatör devresi



# $V_i$ ve $R$ Sabitken



Temel zener regülatör devresi

1. Zener diyotu çıkarın ve zenerin aktif olup olmayacağını tespit edin.
2. İstenilenleri çözerek bulun.
3. Uygunluğunu kıyaslayın.

$$V = V_L = \frac{R_L V_i}{R + R_L}$$

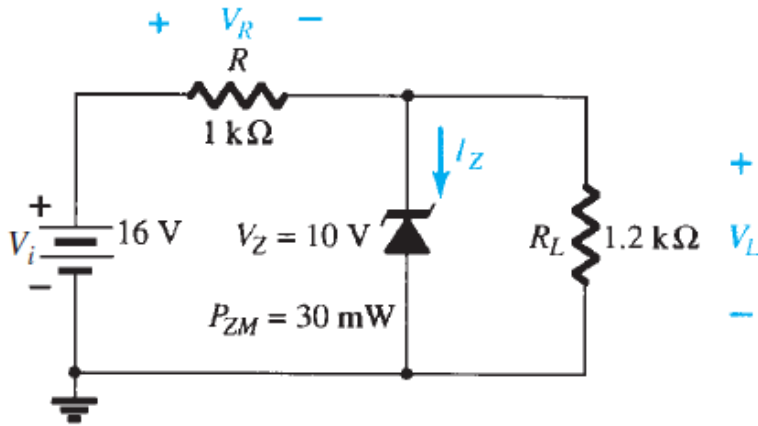
$$V_L = V_Z$$

$$I_R = I_Z + I_L$$

$$I_Z = I_R - I_L$$

$$P_Z = V_Z I_Z$$

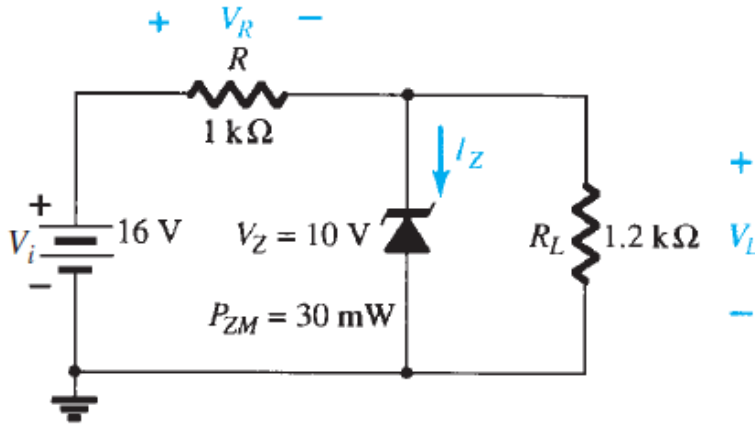
# Örnek



Verilen devre için  $V_L$ ,  $V_R$ ,  $I_Z$  ve  $P_Z$  değerlerini hesaplayın.

Aynı işlemleri  $R_L = 3\text{ k}\Omega$  için tekrarlayın.

# Örnek



Verilen devre için  $V_L$ ,  $V_R$ ,  $I_Z$  ve  $P_Z$  değerlerini hesaplayın.

Aynı işlemleri  $R_L = 3\text{ k}\Omega$  için tekrarlayın.

$$V = \frac{R_L V_i}{R + R_L} = \frac{1.2\text{ k}\Omega (16\text{ V})}{1\text{ k}\Omega + 1.2\text{ k}\Omega} = 8.73\text{ V}$$

$V < V_Z$  olduğu için zener aktif değildir.

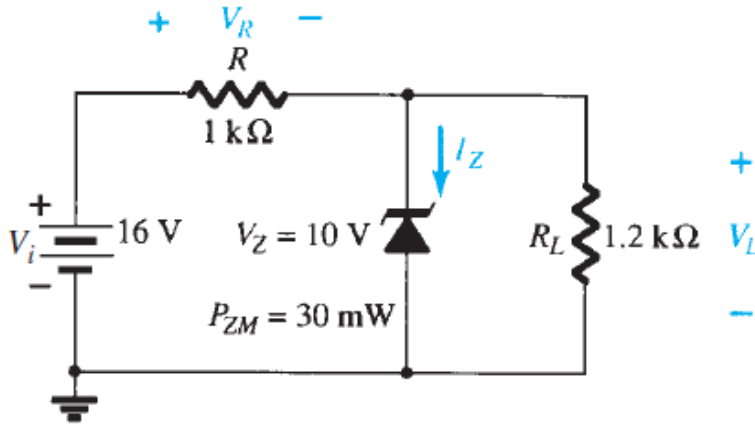
$$V_L = V = 8.73\text{ V}$$

$$V_R = V_i - V_L = 16\text{ V} - 8.73\text{ V} = 7.27\text{ V}$$

$$I_Z = 0\text{ A}$$

$$P_Z = V_Z I_Z = V_Z (0\text{ A}) = 0\text{ W}$$

# Örnek



Verilen devre için  $V_L$ ,  $V_R$ ,  $I_Z$  ve  $P_Z$  değerlerini hesaplayın.

Aynı işlemleri  $R_L=3k$  için tekrarlayın.

$$V = \frac{R_L V_i}{R + R_L} = \frac{3 \text{ k}\Omega (16 \text{ V})}{1 \text{ k}\Omega + 3 \text{ k}\Omega} = 12 \text{ V}$$

$V > V_Z$  olduğu için zener aktiftir.

$$V_L = V_Z = 10 \text{ V}$$

$$V_R = V_i - V_L = 16 \text{ V} - 10 \text{ V} = 6 \text{ V}$$

$$I_L = \frac{V_L}{R_L} = \frac{10 \text{ V}}{3 \text{ k}\Omega} = 3.33 \text{ mA}$$

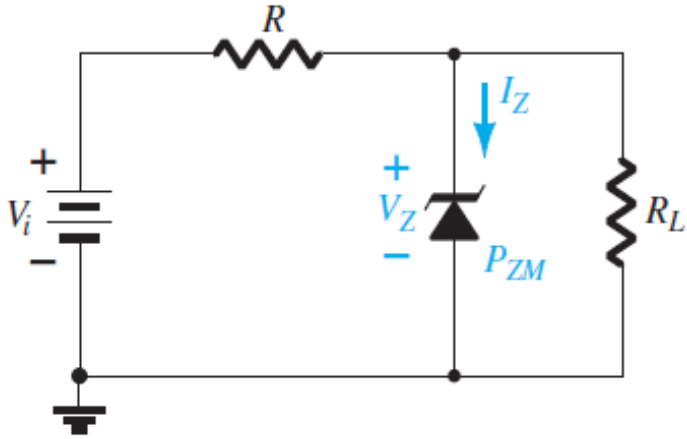
$$I_R = \frac{V_R}{R} = \frac{6 \text{ V}}{1 \text{ k}\Omega} = 6 \text{ mA}$$

$$\begin{aligned} I_Z &= I_R - I_L \text{ [Eq. (2.18)]} \\ &= 6 \text{ mA} - 3.33 \text{ mA} \\ &= 2.67 \text{ mA} \end{aligned}$$

$$P_Z = V_Z I_Z = (10 \text{ V})(2.67 \text{ mA}) = 26.7 \text{ mW}$$

$P_Z < P_{Z_{\max}}$  olduğu için devre uygun şekilde çalışabilir.

# Vi ve R Sabitken



Temel zener regülatör devresi

$$V_L = V_Z = \frac{R_L V_i}{R_L + R}$$

$$R_{L_{\min}} = \frac{R V_Z}{V_i - V_Z}$$

$$I_{L_{\max}} = \frac{V_L}{R_L} = \frac{V_Z}{R_{L_{\min}}}$$

$$V_R = V_i - V_Z$$

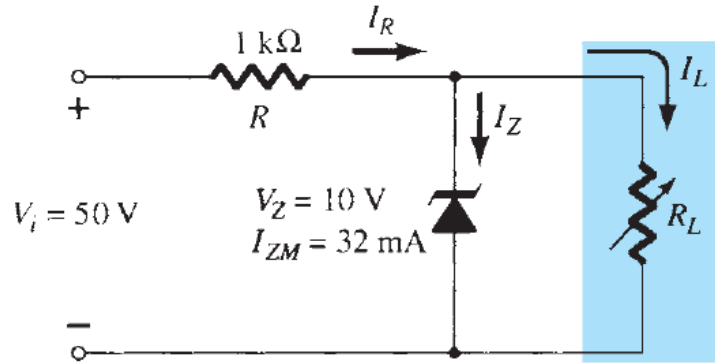
$$I_R = \frac{V_R}{R}$$

$$I_Z = I_R - I_L$$

$$I_{L_{\min}} = I_R - I_{ZM}$$

$$R_{L_{\max}} = \frac{V_Z}{I_{L_{\min}}}$$

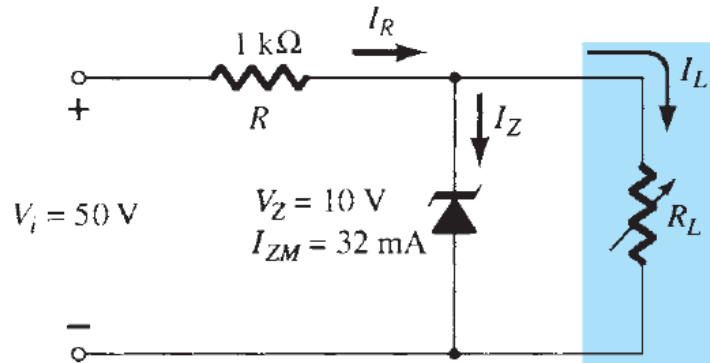
# Örnek



Verilen devrede  $I_L$ , ve  $R_L$  değerlerinin  $V_{RL} = 10\text{ V}$  olacağı durumdaki aralıklarını bulunuz.

Diyotta harcanabilecek max. gücü hesaplayınız.

# Örnek



Verilen devrede  $I_L$  ve  $R_L$  değerlerinin  $V_{RL} = 10 \text{ V}$  olacağı durumdaki aralıklarını bulunuz.

Diyotta harcanabilecek max. gücü hesaplayınız.

$$R_{L_{\min}} = \frac{RV_Z}{V_i - V_Z} = \frac{(1 \text{ k}\Omega)(10 \text{ V})}{50 \text{ V} - 10 \text{ V}} = \frac{10 \text{ k}\Omega}{40} = 250 \Omega$$

$$V_R = V_i - V_Z = 50 \text{ V} - 10 \text{ V} = 40 \text{ V}$$

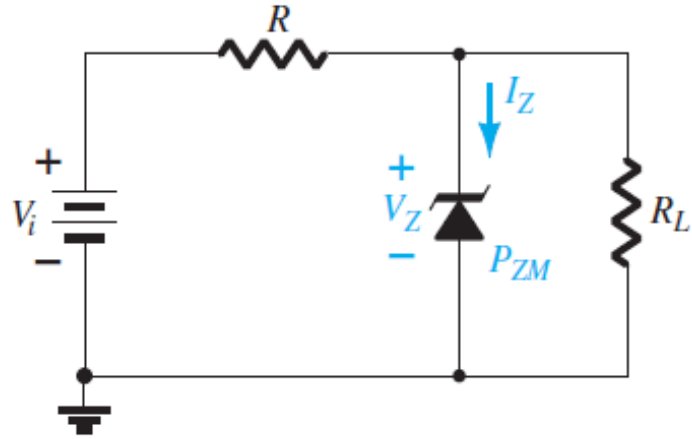
$$I_R = \frac{V_R}{R} = \frac{40 \text{ V}}{1 \text{ k}\Omega} = 40 \text{ mA}$$

$$I_{L_{\min}} = I_R - I_{ZM} = 40 \text{ mA} - 32 \text{ mA} = 8 \text{ mA}$$

$$R_{L_{\max}} = \frac{V_Z}{I_{L_{\min}}} = \frac{10 \text{ V}}{8 \text{ mA}} = 1.25 \text{ k}\Omega$$

$$\begin{aligned} P_{\max} &= V_Z I_{ZM} \\ &= (10 \text{ V})(32 \text{ mA}) = 320 \text{ mW} \end{aligned}$$

# RL Sabit, Vi Değişken



Temel zener regülatör devresi

$$V_L = V_Z = \frac{R_L V_i}{R_L + R}$$

$$V_{i_{\min}} = \frac{(R_L + R)V_Z}{R_L}$$

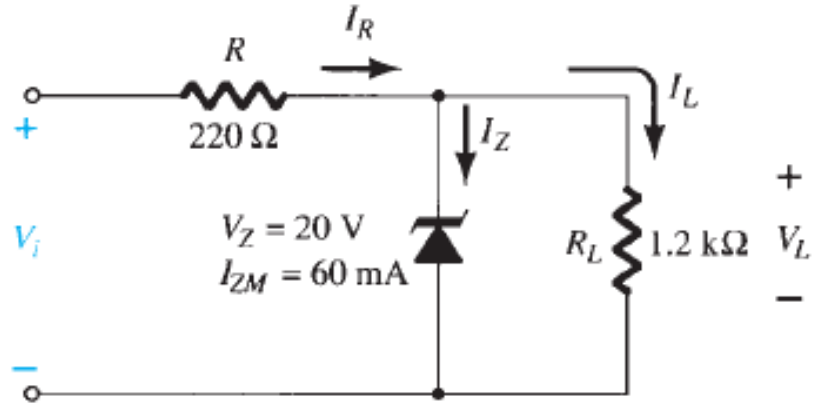
$$I_{R_{\max}} = I_{ZM} + I_L$$

$$V_{i_{\max}} = V_{R_{\max}} + V_Z$$

$$V_{i_{\max}} = I_{R_{\max}} R + V_Z$$

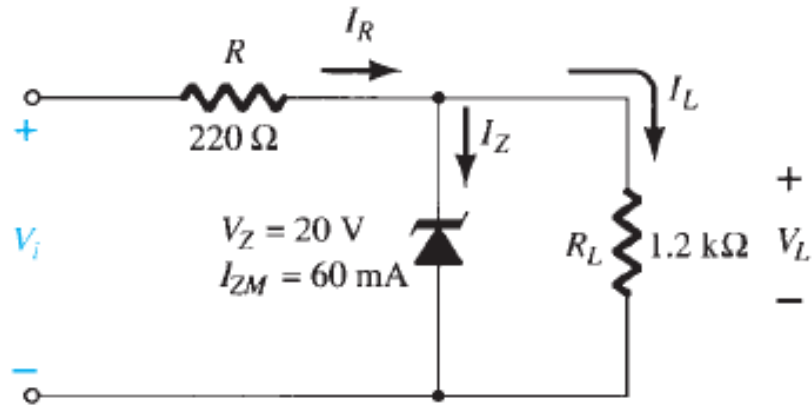


# Örnek



Zener diyotun aktif olarak çalışacağı  $V_i$  aralığını bulunuz.

# Örnek



Zener diyotun aktif olarak çalışacağı  $V_i$  aralığını bulunuz.

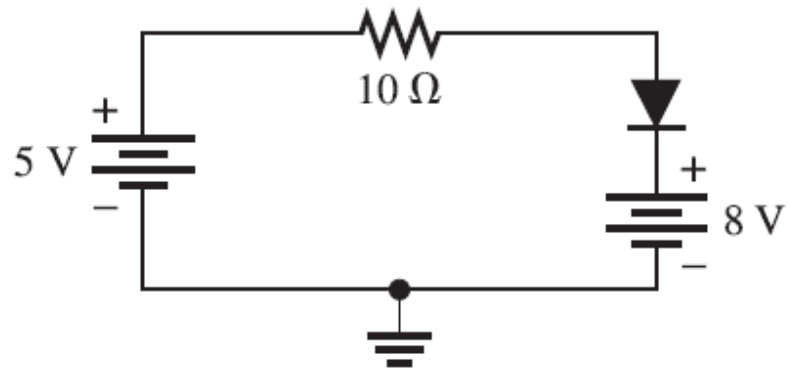
$$V_{i_{\min}} = \frac{(R_L + R)V_Z}{R_L} = \frac{(1200\ \Omega + 220\ \Omega)(20\ \text{V})}{1200\ \Omega} = 23.67\ \text{V}$$

$$I_L = \frac{V_L}{R_L} = \frac{V_Z}{R_L} = \frac{20\ \text{V}}{1.2\ \text{k}\Omega} = 16.67\ \text{mA}$$

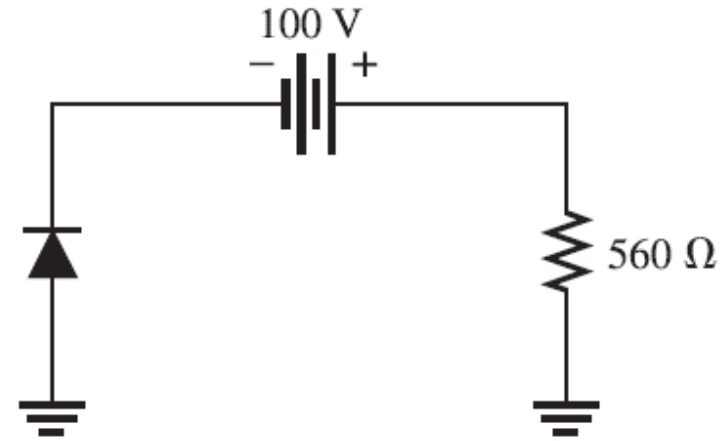
$$\begin{aligned} I_{R_{\max}} &= I_{ZM} + I_L = 60\ \text{mA} + 16.67\ \text{mA} \\ &= 76.67\ \text{mA} \end{aligned}$$

$$\begin{aligned} V_{i_{\max}} &= I_{R_{\max}}R + V_Z \\ &= (76.67\ \text{mA})(0.22\ \text{k}\Omega) + 20\ \text{V} \\ &= 16.87\ \text{V} + 20\ \text{V} \\ &= 36.87\ \text{V} \end{aligned}$$

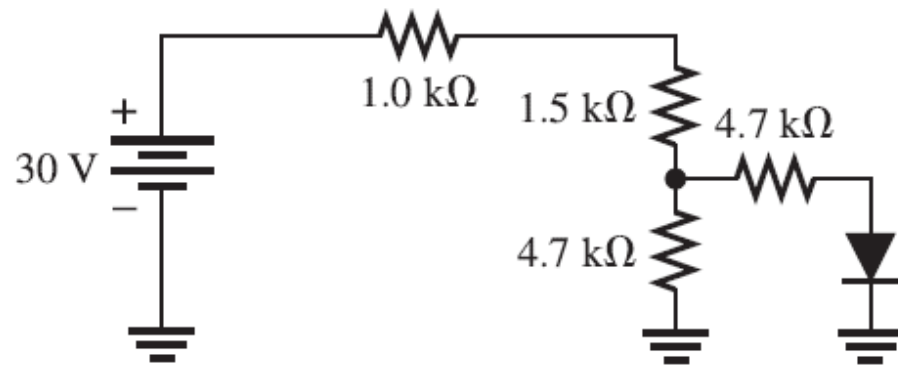
# Örnek 1



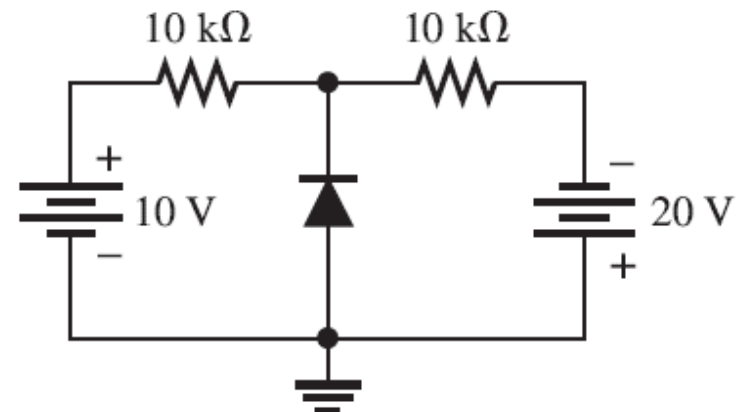
(a)



(b)

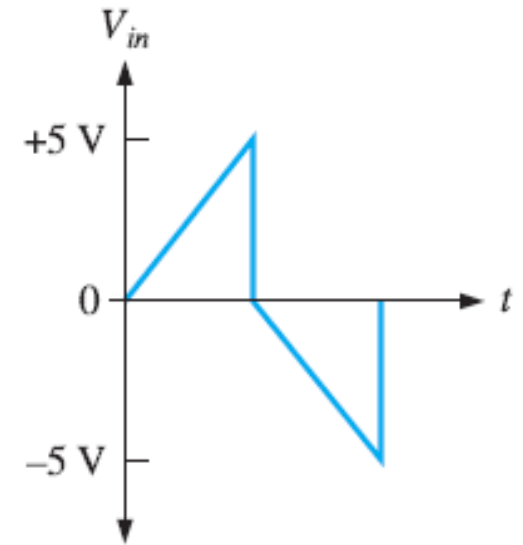
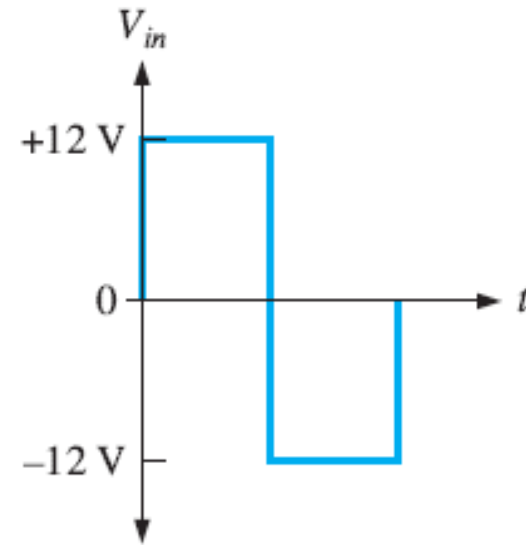
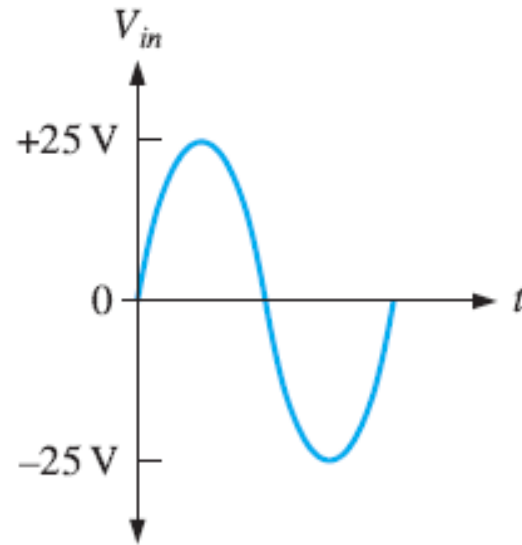
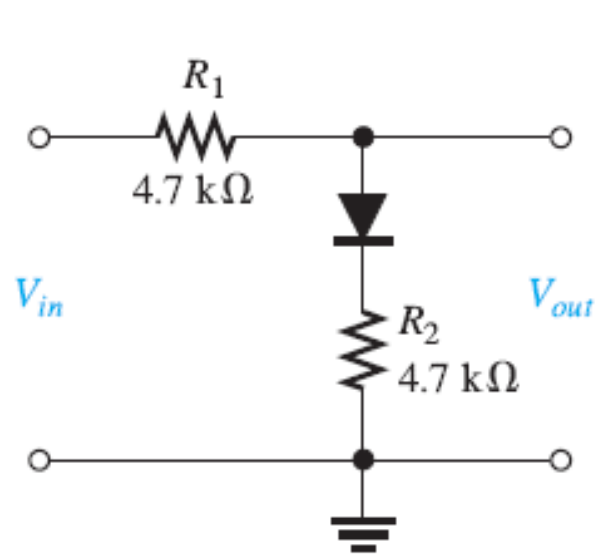


(c)

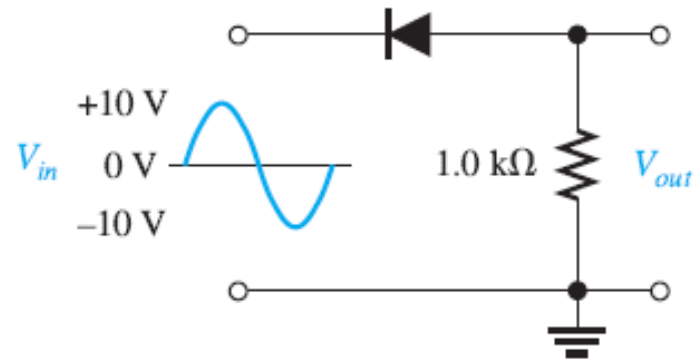


(d)

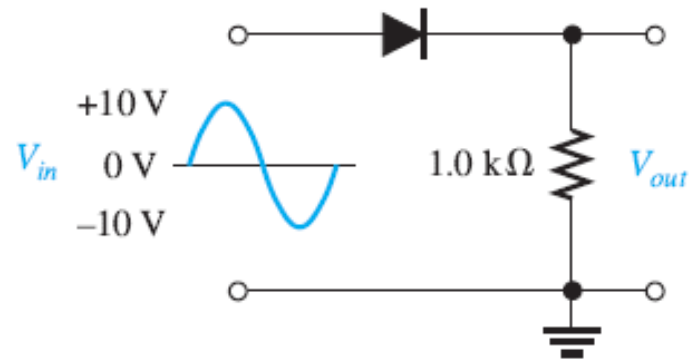
## Örnek 2



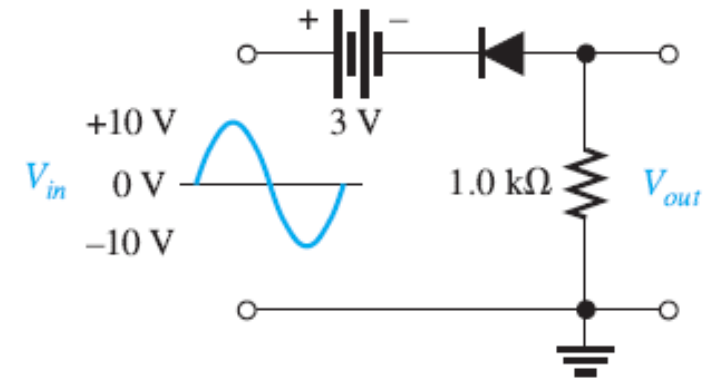
# Örnek 3



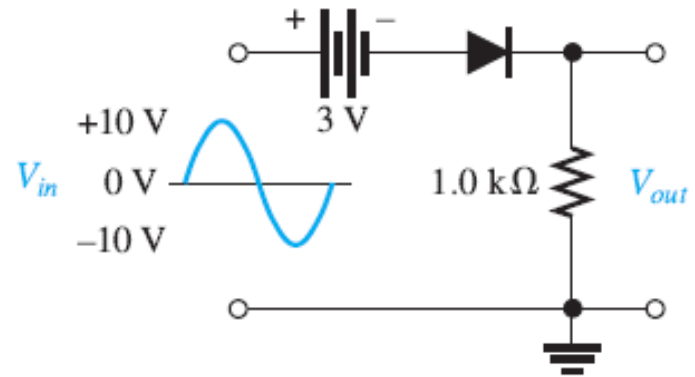
(a)



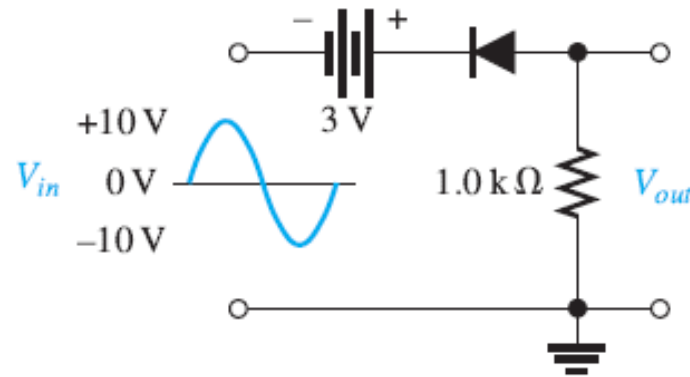
(b)



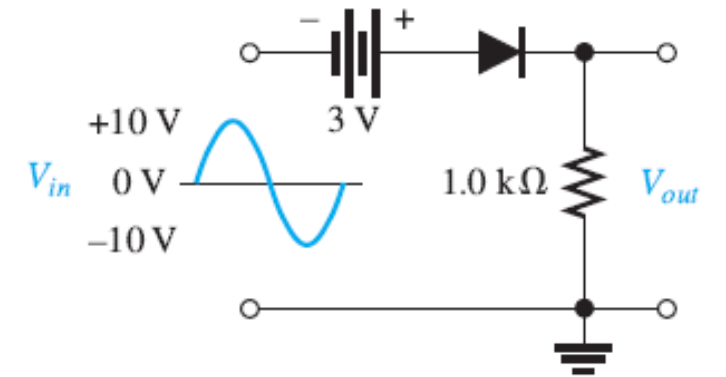
(c)



(d)

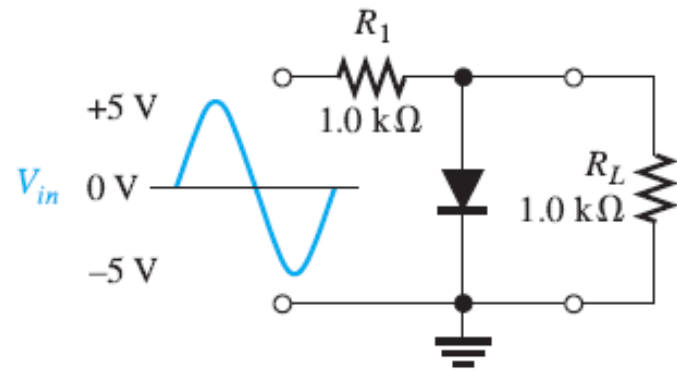


(e)

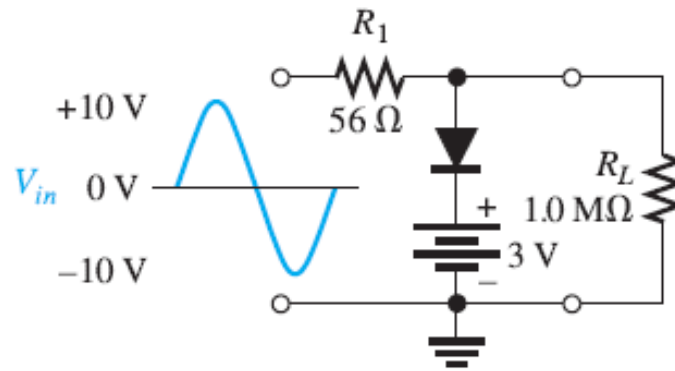


(f)

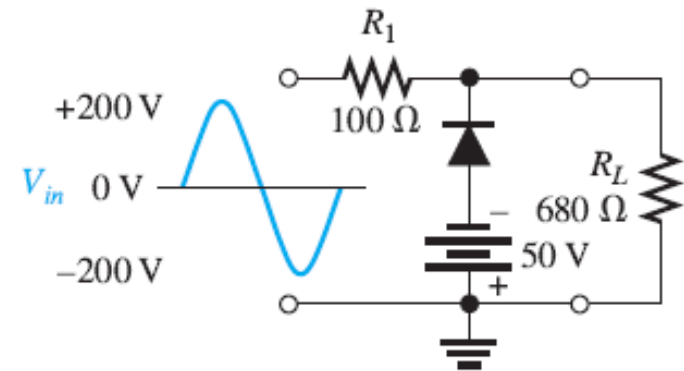
# Örnek 4



(a)

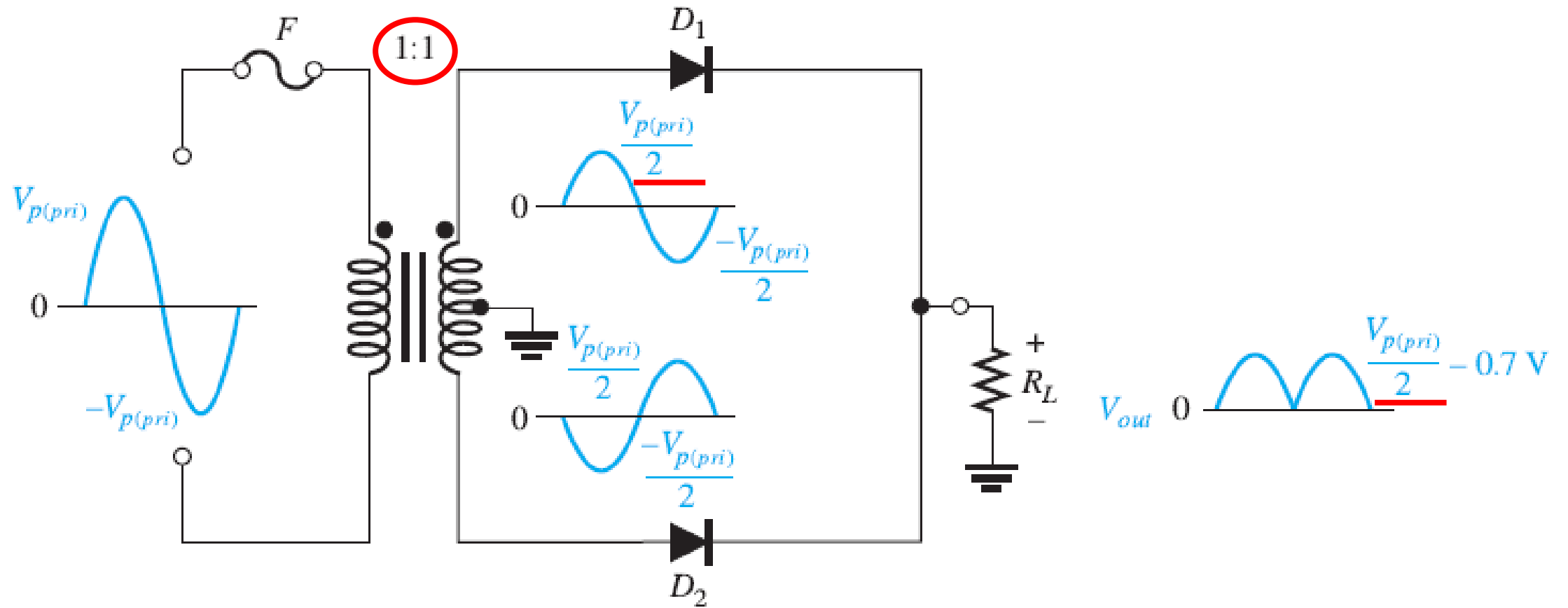


(b)

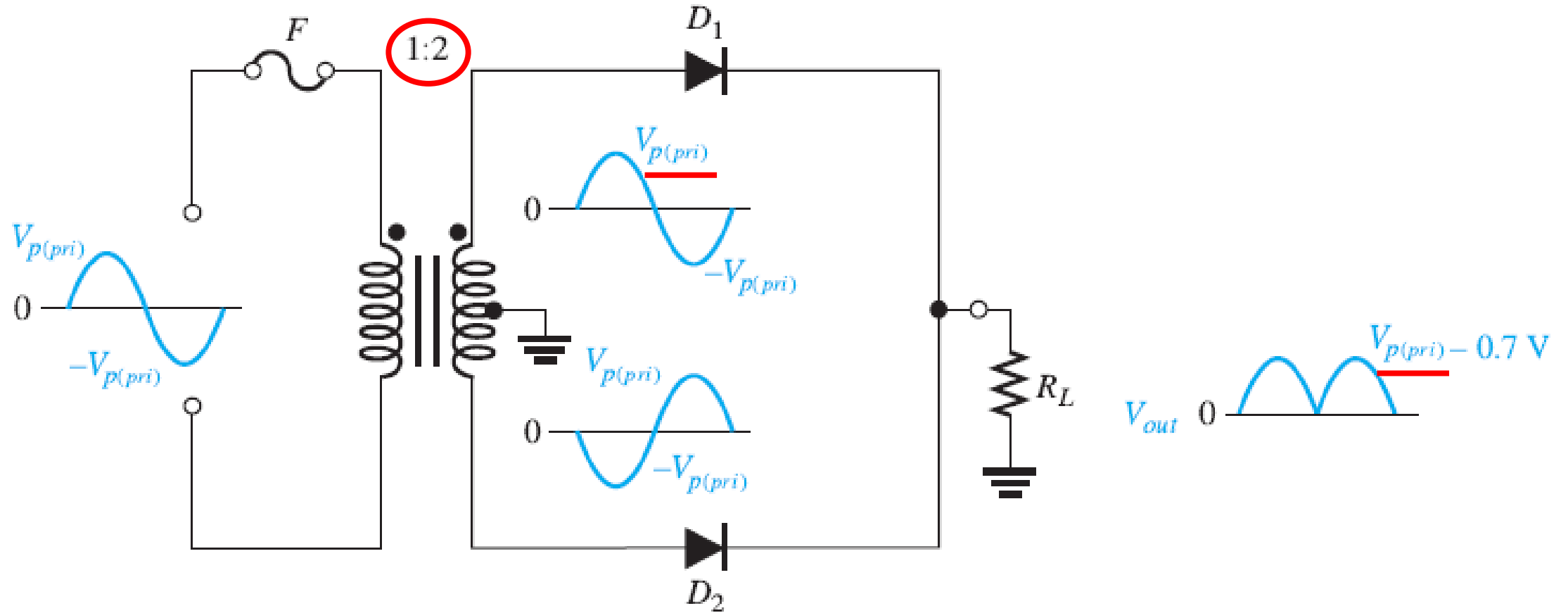


(c)

# Hatırlatma



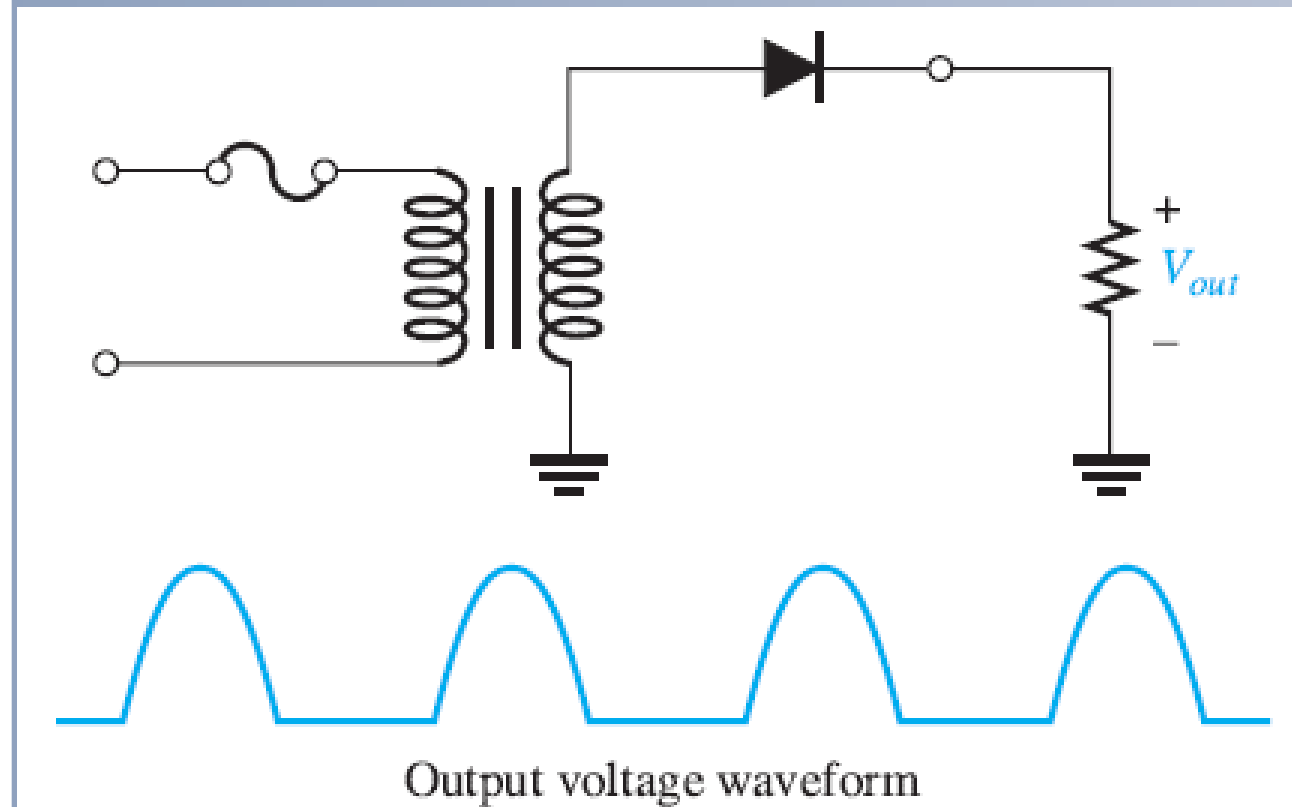
# Hatırlatma





# Hatırlatma

## HALF-WAVE RECTIFIER



- Peak value of output:

$$V_{p(out)} = V_{p(sec)} - 0.7 \text{ V}$$

- Average value of output:

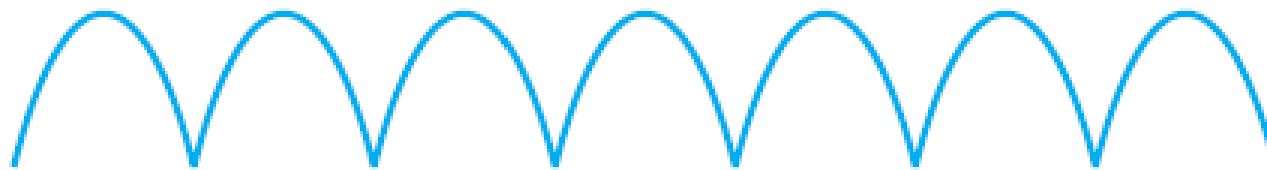
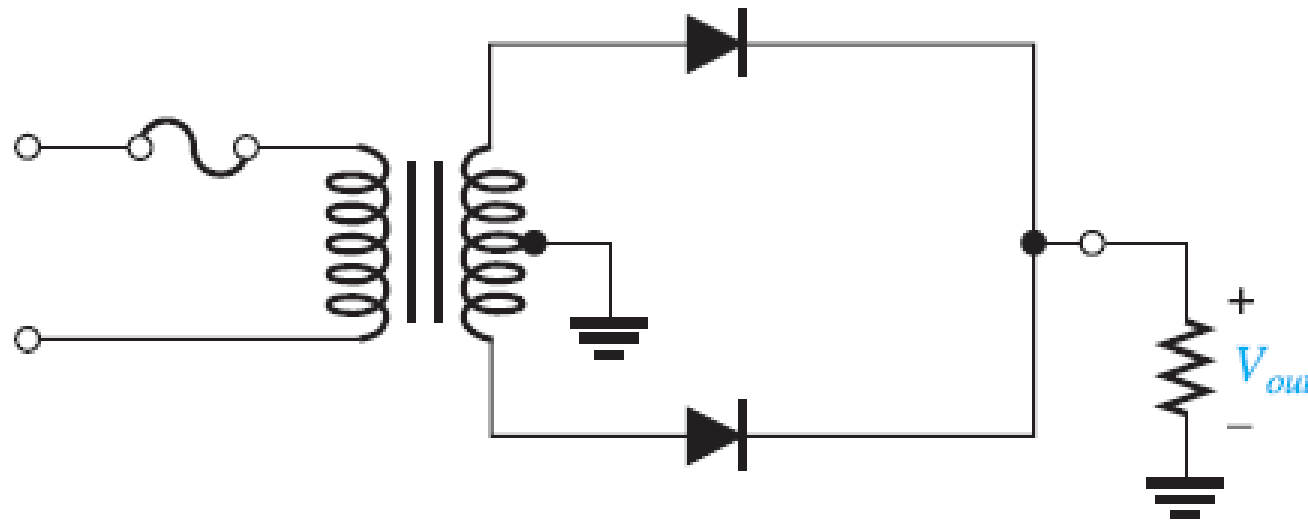
$$V_{AVG} = \frac{V_{p(out)}}{\pi}$$

- Diode peak inverse voltage:

$$PIV = V_{p(sec)}$$

# Hatırlatma

## CENTER-TAPPED FULL-WAVE RECTIFIER



Output voltage waveform

- Peak value of output:

$$V_{p(out)} = \frac{V_{p(sec)}}{2} - 0.7 \text{ V}$$

- Average value of output:

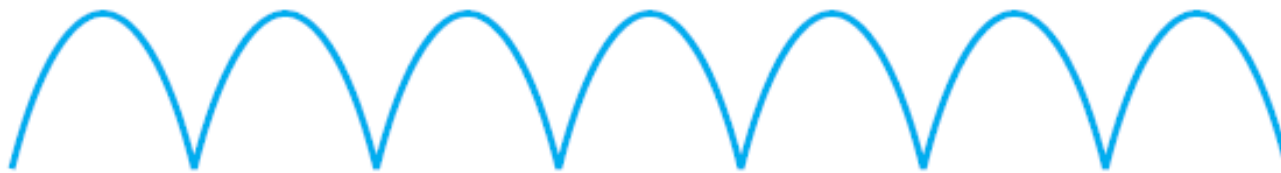
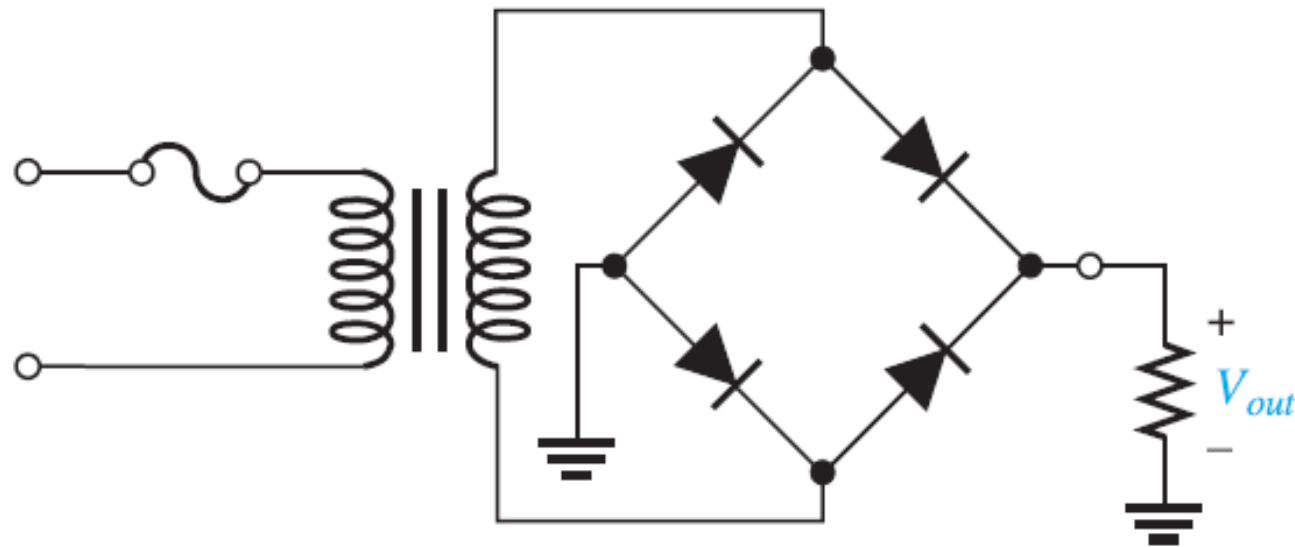
$$V_{AVG} = \frac{2V_{p(out)}}{\pi}$$

- Diode peak inverse voltage:

$$PIV = 2V_{p(out)} + 0.7 \text{ V}$$

# Hatırlatma

## BRIDGE FULL-WAVE RECTIFIER



Output voltage waveform

- Peak value of output:

$$V_{p(out)} = V_{p(sec)} - 1.4 \text{ V}$$

- Average value of output:

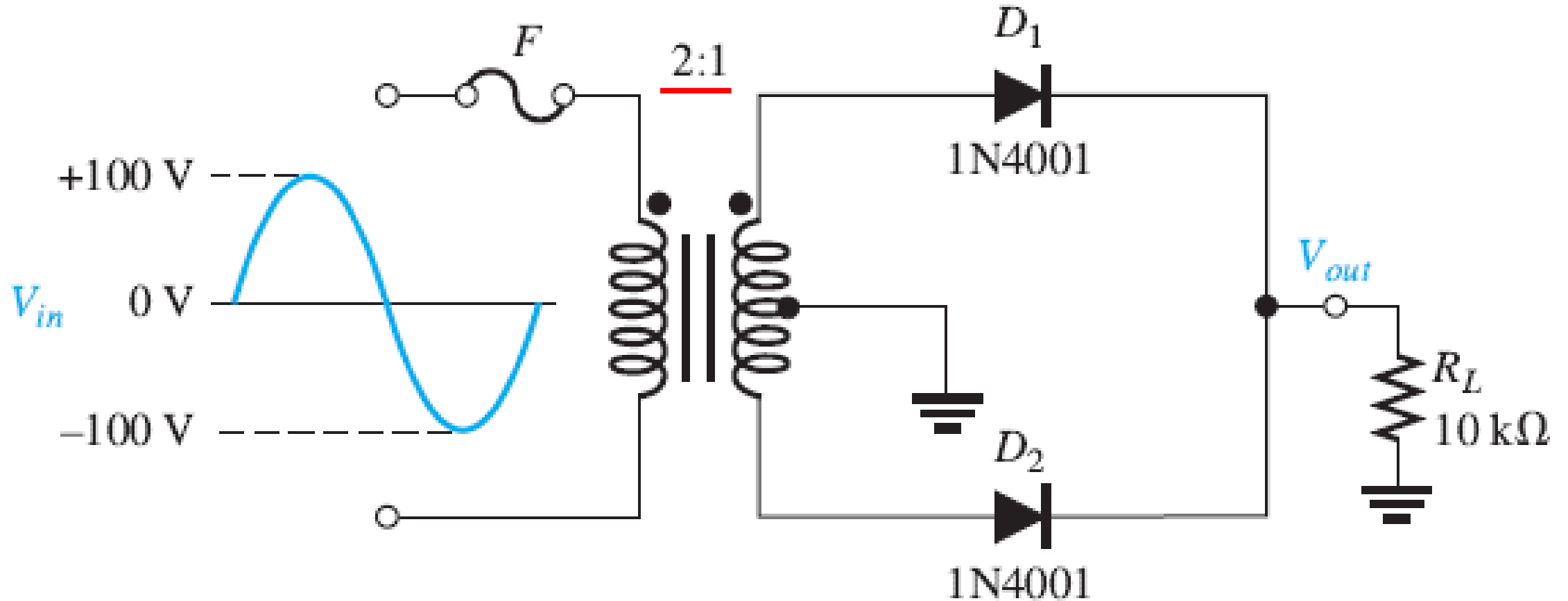
$$V_{AVG} = \frac{2V_{p(out)}}{\pi}$$

- Diode peak inverse voltage:

$$PIV = V_{p(out)} + 0.7 \text{ V}$$

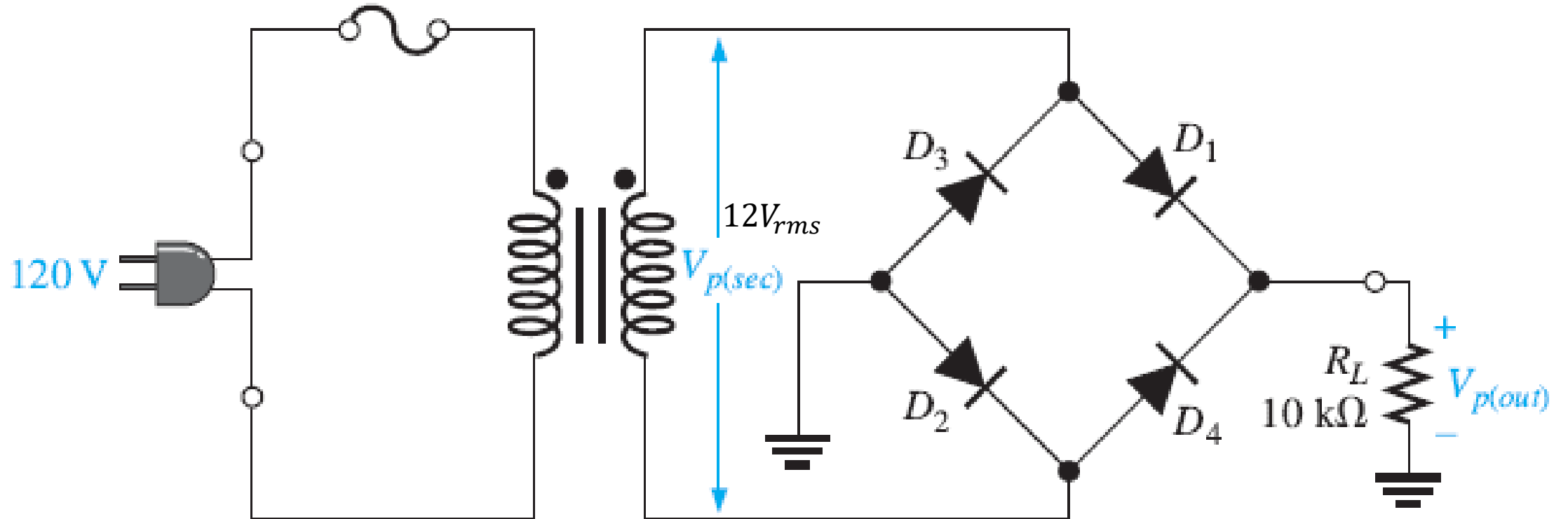
## Örnek 5

Sekonder gerilimini, ters tepe gerilimini ve çıkış gerilimini hesaplayınız.

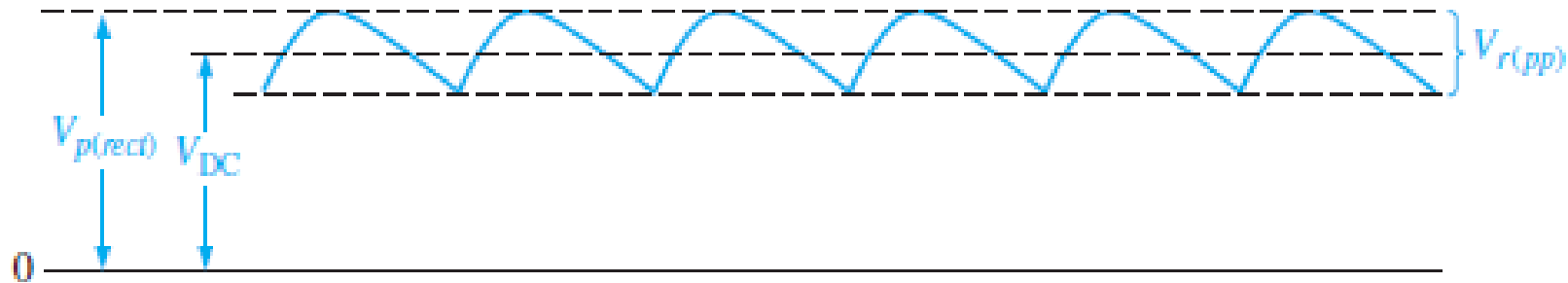


## Örnek 6

Sekonder gerilim  $12V_{rms}$  olması için, transformatör sekonder gerilimini, ters tepe gerilimini ve çıkış gerilimini hesaplayınız.



# Hatırlatma



$$V_{r(pp)} \cong \left( \frac{1}{fR_L C} \right) V_{p(rect)} \quad V_r = \frac{V_{r(pp)}}{2}$$

$$V_{DC} \cong \left( 1 - \frac{1}{2fR_L C} \right) V_{p(rect)}$$

$$V_{rms} = \frac{V_r}{\sqrt{3}} = \frac{V_{r(pp)}}{2\sqrt{3}} \quad r = \frac{V_{rms}}{V_{DC}}$$

# Kaynaklar

- Robert Boylestad and Louis Nashelski,  
*Elektronik Cihazlar ve Devre Teorisi, Palme Yayıncılık*
- Mehmet Akbaba,  
*Elektronik Devreler Ders Notları*
- Thomas L. Floyd  
*Electronic Devices, Merrill Publishin Company*